TELIPORT
Script Your Future
Season 4

TATA
TATA ELXSI

Case Study 2

Team Details

THINK. BUILD. CREATE. WIN

TELIPORT
Script Your Future Season 4

Test a battery before you build it

VOLT-FORGE connects every battery design decision to what it physically changes

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PROBLEM UNDERSTANDING

Cells never share the load evenly

Say you need a Raspberry Pi 5 gateway to survive a **10-minute power cut at 25 W**. Four cells in parallel sounds easy, until the cell closest to the output takes more current than the rest, runs hotter and sets the limit for the whole pack.

DESIRABILITY

You shouldn't need a lab to learn that

Right now you find out by buying parts and hoping. A student can't afford that, and an engineer can't wait for it. Here you wire the circuit on screen, run it and see which cell struggles **before spending a rupee**.

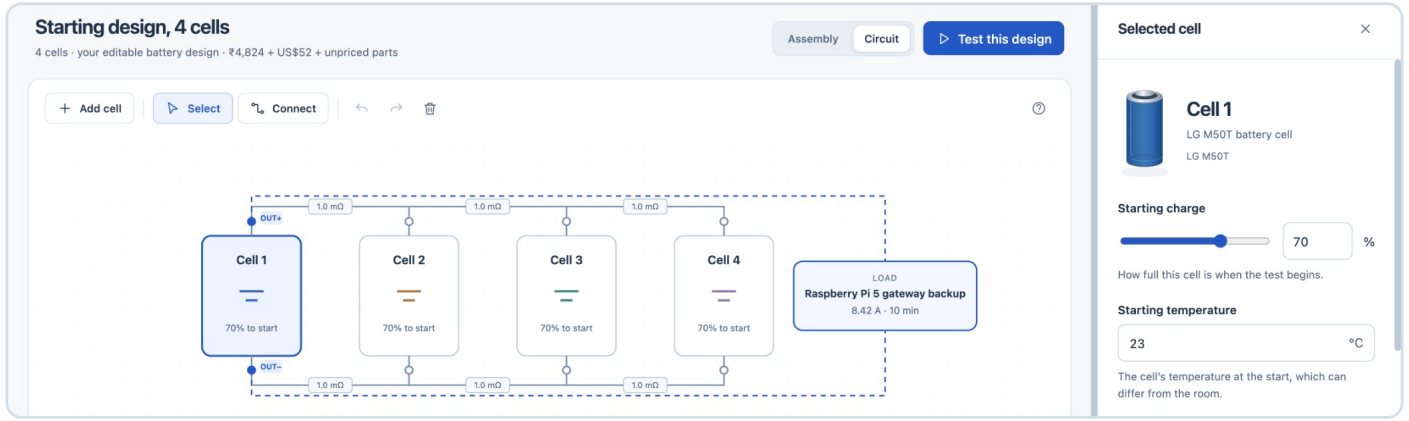
TECHNICAL FEASIBILITY

So how few cells can do this job?

We tested it. **Two cells** kept the Pi running for the full 10 minutes in all 33 tests, even with one weak cell, and never hit their limit. One cell couldn't: the app refused it before it started. And the whole thing runs on a normal laptop.

PROTOTYPE & EVIDENCE

The real Pi design, then 4 cells against 2 under the same load





Result	First test	Second test	Difference
Cells	4	2	−2
Listed cost, priced parts	₹4,824 + US$52	₹2,412 + US$52	
Cell mass	280 g	140 g	−140.00 g
Margin above voltage target	0.40 V	0.24 V	−0.16 V
Warmest cell	24.55 °C	27.47 °C	+2.92 °C

Result	First test	Second test	Difference
Largest cell current	2.64 A	4.37 A	+1.73 A
Energy delivered	5.27 Wh	5.10 Wh	−0.17 Wh
Lowest battery voltage	3.70 V	3.54 V	−0.16 V
Temperature difference	0.40 °C	0.21 °C	−0.19 °C

4,006 what-if runs. Weak cells, bad links, poor cooling: we tried to break it first.

54 lab experiments. Real four-cell test data, with all 194 files checked one by one.

9 tests kept sealed. Held back from every fit, so the final exam uses unseen data.

ENGINEERING QUALITY

Now, why should you trust that?

Heat is just electricity that never reached the load. So every run has to balance: energy out of the cells equals energy delivered plus heat. If it doesn't, the result gets flagged. Cooling varies by rig, so temperature stays a **range**.

INNOVATION

You don't have to guess the answer

You could try every cell count by hand. Or press **Find the smallest design**: it rebuilds the circuit at each size, tests them all and shows what survives. For the Pi, that meant 2 cells, not 4, at half the cell cost.

BUSINESS VIABILITY



41.3 lakh TAM, engineering students

15.6 lakh SAM, electrical, electronics, mechanical

130 SOM, colleges, 5% assumed

₹35 crore

to build all of it, from a measured dataset to the full app. 100 person-years, about 2% of AspenTech's yearly R&D.

These results are simulated. Next we bench-test real cells and put it in front of students and engineers.